04. Preprocessing & Feature Engineering

Reconstruct the notebook's transformations and their modelling implications.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

Exact transformation sequence

Step	Notebook implementation	Result
Missing income	Fill Income nulls with median of full dataset	24 values imputed
Age	Age = 2026 - Year_Birth	New age feature
Tenure	Parse Dt_Customer with dayfirst=True; subtract from max join date	Customer_Tenure_Days
Spending	Sum six Mnt_* fields	Total_Spending
Children	Kidhome + Teenhome	Total_Children
Education	Basic/2n Cycle→Undergraduate; Graduation→Graduate; Master/PhD→Postgraduate	3 grouped categories
Living arrangement	Married/Together→Partner; other listed statuses→Alone	Living_With
Drop columns	ID, Year_Birth, Marital_Status, Kidhome, Teenhome, Dt_Customer and six Mnt_* fields	15 columns remain
Filter	Age < 90 and Income < 600,000	2,236 rows
Encoding	OneHotEncoder on Education and Living_With	18 total features
Scaling	StandardScaler fit_transform on all rows	18 standardized columns
PCA	PCA(n_components=3)	3 components

Why these steps are used

- Median imputation is less sensitive than mean imputation to a skewed income distribution.
- Age and tenure convert raw dates into interpretable elapsed-time features.
- Total_Spending summarises monetary engagement, while Total_Children captures household composition.
- One-hot encoding avoids inventing an ordinal relationship between nominal categories.
- StandardScaler prevents large-unit features such as income or spending from dominating Euclidean distances purely because of scale.
- PCA compresses correlated features into orthogonal components that capture as much variance as possible.

Important caveats

- Imputation, encoding, scaling and PCA are fitted on the complete dataset. For a production segmentation pipeline, fit these transformations on a defined reference/training cohort and reuse the fitted objects for new customers.
- The notebook drops the six product-category spending columns after creating Total_Spending. This loses product-mix information; customers with the same total but different baskets become harder to distinguish.
- Response is kept as an input feature. Exclude it if segmentation is intended to precede a future campaign or if it is an outcome the business wants to explain.

- Age uses the hard-coded year 2026. For reproducibility, specify a fixed as-of date and avoid silently changing age across reruns.
- The two income/age filters need business justification and an audit of excluded records.

PCA intuition

PCA finds directions (principal components) that capture maximum variance in the standardized data. Here the first three components explain 23.16% + 11.39% + 10.41% = 44.95% of total variance. Thus 55.05% of standardized feature variance is not represented in the three-dimensional clustering space. PCA is a compression trade-off, not a guarantee that the most commercially useful structure is preserved.

Better pipeline design

Use a fitted preprocessing pipeline: imputer → one-hot encoder / scaler → optional PCA → clustering. Fit it on a reference cohort, validate stability, and apply the same transformations to future customers. Decide deliberately whether to include Response and whether to preserve product-category mix.